

Why joint training may fail to acquire useful scales

Readout learning changes the residual that drives slope learning. The companion brief, *Gamma controls readout learning speed*, establishes why scale matters. Here we ask why joint training often fails to acquire it. Rate interventions support competition between readout and geometry; an early theorem captures signal depletion; the current approach studies the fine-residual sensitivity that survives coarse fitting. The long-time prediction is still open.

Notation. Parameter-vector norms are Euclidean; residual norms use the stated data measure.

a_j, b_j, w_j, d	Hidden slope, hidden bias, readout weight, and output bias.
$\gamma_j = a_j , \bar{\gamma}$	Individual slope magnitude and its population mean.
$r = f - y, L = \ r\ _m^2/2$	Residual and half-MSE; $\ r\ _m^2 = m^{-1} \sum_i r(x_i)^2$.
$\kappa, t = \eta n$	Readout/geometry rate ratio and flow clock for GD update n .
$e_C, e_H, \mu_a^{\text{eff}}$	Coarse/fine residual coefficients and effective slope sensitivity (Section 3).

1. Empirical motivation: changing the race changes gamma

The jointly trained model is

$$f(x) = d + \sum_{j=1}^W w_j \tanh(a_j x + b_j). \quad (1)$$

Reparameterization first motivated the timescale hypothesis: a fixed scaling $\theta = Du$ makes a GD step in u induce $\Delta\theta = -\eta DD^T \nabla_\theta L$. For example, training signed slopes in grid units ha_j , with fixed reference spacing h , multiplies the physical slope step by h^{-2} . These changes alter the relative mobility of readout and geometry. The subsequent direct intervention isolates the rate ratio: keep slope and hidden-bias rate $\eta = 0.002$ fixed, and vary the readout and output-bias rate $\kappa\eta$, from exactly the same random initial parameters [1].

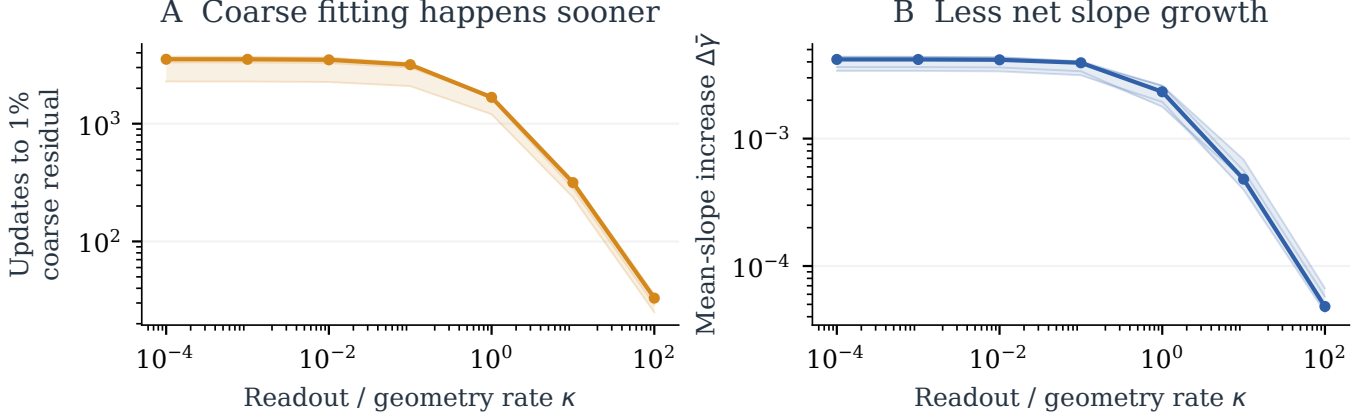


Figure 1: Faster readout removes coarse error sooner and leaves less slope growth. Ordinary simultaneous GD, $W = 177, 2,048$ training midpoints, five paired seeds, 20,000 updates. Target: $0.3\phi_0 + 0.4\phi_1 + \sqrt{0.75}\phi_9$, with empirically orthonormal polynomial modes ϕ_k . Coarse means constant and linear. Bold lines show medians; faint curves and shading retain all seeds. B measures net mean-slope change at the common final update. These are allocation effects while fine error remains large.

Increasing κ from 1 to 100 changes the median coarse-fitting time from 1,675 to 33 updates and mean slope growth from 0.002332 to 0.00004810: **48.5 times less growth**, with independent-grid MSE still about 0.75. At the same coarse-error threshold, the growth contrast is already present (0.002298 versus 0.00004822). It is therefore not just a comparison at different stages of the transient. The effect is target dependent: Runge can gain slope growth with faster readout. The evidence supports a coupled allocation mechanism, not a universal prescription to slow the readout.

2. First proof idea: quantify early signal depletion

The first proof asks whether readout fitting removes an initially available slope signal while appreciable error remains. A small gradient alone does not answer this: it might simply reflect successful fitting. We therefore study

$$R(t) = \|r(t)\|, \quad g_a = \nabla_a L, \quad \Xi(t) = \frac{\|g_a(t)\|}{R(t)}. \quad (2)$$

This is the current slope gradient per unit residual, including any signal regenerated during training [2].

The theorem’s setting. Use equal-rate gradient flow, continuous $L^2(dx/2)$ loss on $[-1, 1]$, no output bias, and *zero initial readout*. Hidden slopes and biases are independent uniforms on $[-\sqrt{6/(W+1)}, \sqrt{6/(W+1)}]$. The fixed target y is centered, normalized, and entire, with $\beta = 3\langle y, x \rangle \neq 0$ and $\|y - \beta x\| > 0$. For each fixed $H_0 > 3$, at sufficiently large width and with arbitrarily high fixed probability, signal first forms at a width-independent time t_f , then obeys

$$\Xi(t_f) \geq s_y > 0, \quad \Xi(t) \leq C_{y, H_0}(e^{-t/3} + W^{-1}), \quad 0 \leq t \leq H_0 \log W. \quad (3)$$

In particular, $\Xi = O(W^{-1})$ throughout $[3 \log W, H_0 \log W]$, even though the target has a non-affine component.

Why the proof works. At small preactivations, $\tanh$ is approximately affine. Its exact slope gradient can be written

$$(g_a)_j = w_j \langle r, x \rangle - w_j \langle r, x \tanh^2(a_j x + b_j) \rangle. \quad (4)$$

The coupled affine reference describes fitting the constant and linear moments. A parameter comparison and nonlinear remainder estimates bound both their regeneration and the second term in (4). Thus the theorem bounds the total surviving signal, rather than merely showing that the initial coarse residual disappeared.

Where the readout enters. Let $q = (a, b)$ and let H_{aI} be the corresponding loss-Hessian block. Where $\|g_a\|, R > 0$, equal-rate flow gives the exact identity

$$-\frac{d}{dt} \log \Xi = D_w + D_q, \quad D_I = \frac{g_a^T H_{aI} g_a}{\|g_a\|^2} - \frac{\|g_I\|^2}{R^2}. \quad (5)$$

On the shorter attribution window from t_f to $t_f + \rho \log W / \nu_y$, with $0 < \rho < 1/2$ and $\nu_y = 2\sqrt{1 + \beta^2}/3$, the readout’s share of the leading logarithmic decline tends in probability to

$$\pi_y = \frac{1}{2} + \frac{1}{2\sqrt{1 + \beta^2}} \in (3/4, 1). \quad (6)$$

This is a signed attribution along the joint trajectory. It says more than three quarters of that leading decline is attributable to readout adaptation; it is not three quarters of all possible gamma growth, or a frozen-readout counterfactual. Individual contributions can later become negative.

Why this was not yet quantitatively predictive. The proof supplies width and time scaling, but its conservative constants do not give tight bounds at our experimental widths. For one audited target an explicit old bound contains $905520/W$, subject to sufficient-width conditions; the refined asymptotic post-depletion prediction is about $0.2623/W$. These are not competing certified bounds at $W = 512$: the contrast exposes the slack between a uniform proof estimate and the predicted leading signal. The guarantee also covers an early logarithmic-time window and a zero-readout law, whereas the long D34 trajectories start with random nonzero readouts. Accurate cubic or quintic reference forecasts help, but measured agreement does not certify a uniformly small error envelope.

The proof establishes an early mechanism. The next task is to explain the force that remains, including its subsequent recovery, on the actual training trajectory. That motivates resolving the gradient-flow equations more directly.

3. Current approach: the sensitivity left after coarse fitting

Return to the empirical model (1), with all four blocks trained, random nonzero readouts, and equal physical rates. Its slope equation is

$$\dot{a}_j = -(g_a)_j = -w_j \langle r(x) x \operatorname{sech}^2(a_j x + b_j) \rangle_m. \quad (7)$$

Residual size alone cannot determine this velocity: the residual must align with each current slope tangent, and the readout weight multiplies that tangent. We now identify the force after constant and linear fitting have relaxed [3].

Resolve the residual into coarse and fine components. Choose a fixed empirical orthonormal basis $q_k(x)$ on the training points, with $q_0 = 1$, $q_1 = x/\sqrt{\langle x^2 \rangle_m}$ on the symmetric grid. Set $e_k = \langle q_k r \rangle_m$, $e_C = (e_0, e_1)^T$, and let e_H contain the *complete* remaining complement. Define the modal Jacobian block $(J_I)_{kj} = \langle q_k \partial f / \partial \theta_{I,j} \rangle_m$ for $I \in \{a, b, w, d\}$. Then $g_I = J_I^T e$, and equal-rate flow gives

$$\dot{e} = -K e, \quad K = \sum_I J_I J_I^T, \quad \begin{aligned} \dot{e}_C &= -K_{CC} e_C - K_{CH} e_H, \\ \dot{e}_H &= -K_{HC} e_C - K_{HH} e_H. \end{aligned} \quad (8)$$

All parameter blocks contribute to K . The coarse/fine split concerns output error, rather than a split between readout and geometry parameters.

Coarse equilibrium is generally nonzero. If K_{CC} is invertible, hold its current blocks and e_H fixed to find the instantaneous coarse balance:

$$B = K_{CC}^{-1} K_{CH}, \quad e_C^{\text{bal}} = -B e_H, \quad z_C = e_C + B e_H. \quad (9)$$

Here z_C measures disagreement with that moving balance. Substituting $e_C = -B e_H + z_C$ into the slope gradient and fine-residual equation gives

$$\boxed{g_a = J_{a,C}^T z_C + T_a e_H, \quad \dot{e}_H = -S e_H - K_{HC} z_C}, \quad (10)$$

$$T_a = J_{a,H}^T - J_{a,C}^T B, \quad S = K_{HH} - K_{HC} B. \quad (11)$$

The first force in (10) comes from imperfect tracking; the second survives at coarse balance. Crucially, $T_a e_H$ includes the balanced coarse response $-J_{a,C}^T B e_H$. Setting $e_C = 0$ would omit it. “Fine” also includes generated low-degree nonlinear errors, not only the hardest target mode. These identities are exact for the full complement; a truncated modal calculation must retain its omitted-residual corrections.

The measurements identify which force to study. Across 13 targets and five seeds, updates 20,000–600,000, the tracking channel contributes at most **0.469%** of $\sum_n (\|J_{a,C}^T z_C\| + \|T_a e_H\|)$ in ordinary equal-rate GD [4]. This is a fraction of summed component norms, not net gamma growth. It supports concentrating on $T_a e_H$ over this observed interval. The same simplification does not hold for Adam.

Arriving at the effective sensitivity. For $R_H = \|e_H\| > 0$, let $u_H = e_H / R_H$ and define

$$G_a = T_a^T T_a, \quad \boxed{\mu_a^{\text{eff}} = u_H^T G_a u_H = \frac{\|T_a e_H\|^2}{\|e_H\|^2}, \quad \|T_a e_H\| = R_H \sqrt{\mu_a^{\text{eff}}}}. \quad (12)$$

This is a squared, direction-dependent sensitivity of slopes to the current fine residual. It separates *how much error remains* from *how strongly that error can move the slopes*. It also incorporates interference between direct fine forcing and the induced coarse response.

What this sensitivity suggests next

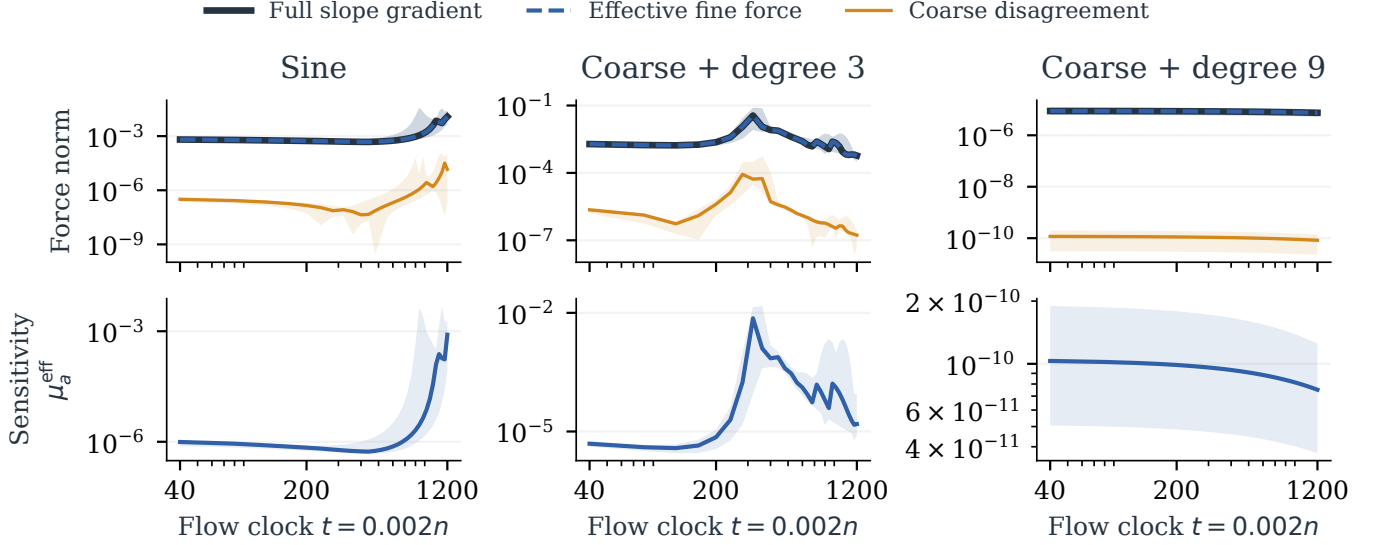


Figure 2: Coarse disagreement is small, but the surviving sensitivity is target dependent. Equal-rate GD, $W = 177$, 2,048 training points, seeds 0–4; the polynomial targets share their coarse coefficients and fine-target energy. Upper row: full slope-gradient norm, effective fine-force norm, and tracking-force norm. Black and dashed blue nearly coincide. Lower row: (12). Lines join saved-state medians; shading spans seeds, with separate vertical ranges. The clock $t = 0.002n$ labels GD states; these are measured trajectories, not continuous-time certificates. Gradients are evaluated at those states.

At update 600,000, the degree-nine case still has median fine-residual norm $R_H = 0.8660$, but $\mu_a^{\text{eff}} = 7.51 \times 10^{-11}$. The degree-three case has $R_H = 0.1596$ and $\mu_a^{\text{eff}} = 1.54 \times 10^{-5}$ [4]. The latter recovers appreciable sensitivity before weakening again; degree nine remains weakly coupled despite its large error. Thus a single small endpoint gradient cannot explain the history.

The race now has an explicit dynamical form. Write $F_a = T_a e_H$. Differentiating the force gives

$$\dot{F}_a = \underbrace{\dot{T}_a e_H}_{\text{changing sensitivity map}} - \underbrace{T_a S e_H}_{\text{evolving fine residual}} - \underbrace{T_a K_{HC} z_C}_{\text{tracking correction}}. \quad (13)$$

Readout evolution affects both T_a and S : it can improve coupling while also fitting away the residual that supplies the drive. Residual direction can rotate as components fit at different speeds, so even decreasing residual energy need not decrease force. These effects explain why permanently discarding nonlinear recovery would miss the competition.

To connect sensitivity to acquisition, define the signed outward alignment $\chi_a = -\text{sign}(a)^T F_a / (\sqrt{W} \|F_a\|)$ where $F_a \neq 0$. Away from slope zero crossings, the exact flow identity is

$$\dot{\gamma} = \frac{R_H \sqrt{\mu_a^{\text{eff}}} \chi_a}{\sqrt{W}} - \frac{\text{sign}(a)^T J_{a,C}^T z_C}{W}, \quad |\chi_a| \leq 1. \quad (14)$$

Sensitivity determines available force; alignment determines whether it sharpens or flattens the network. A population claim additionally needs its distribution across neurons. With bounds on the full slope force, the exact GD path budget $\eta \sum_n \|g_{a,n}\|$ can be compared with the distance to a specified acquired population [3]. The measured budget is a retrospective check; a prediction requires independently justified bounds along the trajectory.

Working hypothesis. After coarse relaxation, readout–geometry feedback can leave the remaining error weakly coupled to slopes, or remove its useful driving components before enough outward motion accumulates. The promising object is therefore the *evolving, target-dependent sensitivity* in (12), together with the force evolution in (13). The next proof must control this feedback and its tracking correction from initialization or a justified entry state. The decomposition and measurements identify that problem; they do not yet prove a long-time acquisition barrier.

Sources and reproduction. [1] D34 rate intervention; [2] Early-depletion theory, 20 September 2026; [3] Coupled-force theory; [4] Expanded GD/Adam force study. The companion motivation is [Gamma controls readout learning speed](#). Figures replot existing fp64 results; no new training was run. Exact inputs, selected plotting values, checks, and hashes are in the [evidence record](#). Rebuild figures with `python -m experiments.expD34_readout_race.pi_brief_figures`; the LaTeX source gives the compilation command. Local source links resolve within the research checkout; the argument above is self-contained.